

Datathon: Hands-on Afternoon Activity

UCSB Data Day – Southern California Consortium for Data Science

2026-03-07

Channel Islands

The **Channel Islands** are a chain of eight islands located off the southern coast of California in the Pacific Ocean. They are known for their stunning natural beauty, ecological richness, and deep cultural history.

The Eight Channel Islands (North to South)

1. San Miguel Island
2. Santa Rosa Island
3. Santa Cruz Island
4. Anacapa Island
5. Santa Barbara Island
6. San Nicolas Island
7. San Clemente Island
8. Santa Catalina Island

Channel Islands National Park

- Includes **five islands**: *San Miguel*, *Santa Rosa*, *Santa Cruz*, *Anacapa*, and *Santa Barbara*.
- Sometimes called the “**Galápagos of North America**” for its rich biodiversity and many **endemic species**.
- Accessible only by **boat or small plane**.
- Ideal for hiking, kayaking, camping, and wildlife observation.

Cultural and Historical Significance

- Originally inhabited by **Chumash and Tongva** indigenous groups for thousands of years.
- Home to important archaeological and cultural heritage sites.

Island Fox (*Urocyon littoralis*)

The **island fox** is a small, charismatic canid species **endemic** to six of the eight **Channel Islands** off the coast of southern California. It’s one of the best examples of **island dwarfism**, having evolved from the mainland gray fox (*Urocyon cinereoargenteus*) to become significantly smaller.

In the **1990s**, several island fox populations faced **near extinction** due to golden eagle predation, canine distemper virus, and habitat degradation. The island fox was listed as endangered in 2004. Recovery efforts of the island fox included removal of golden eagles, vaccination and captive breeding programs, reintroduction of native bald eagles, and habitat restoration.

Activity 1

As a start, you will investigate one of the two sections, Fox Weight or Fox Reproductive Status. Follow the guidelines in the section of your choice and complete one of the challenges. **You may also analyze the data in any other way or work on the other section, but it is not necessary.**

Data and Prep

The data provided is part of an ongoing longitudinal study of island foxes from 2014 to 2016 of trap monitoring. This resulted in collection of 4,975 recorded trap observations, which may have contained a fox or not. Other information collected were Sex, Age Class, Weight, Body Condition, Reproductive Status, and Vaccinations. Information on the data collected can be found in the [Databook](#).

Loading Packages

The tidyverse suite is a collection of data analysis packages:

```
library(tidyverse)

-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v ggplot2    3.5.2      v tibble     3.3.0
v lubridate  1.9.4      v tidyr      1.3.1
v purrr      1.1.0
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

Loading Data

We will load the data into the R workspace using the `read_csv` function:

```
# Need to modify to load the file from the location where you downloaded it
df <- read_csv(here::here("data_for_datathon_2014_2016.csv")) # Loads the Data Set into the object df

Rows: 4975 Columns: 16
-- Column specification -----
Delimiter: ","
chr  (9): Island, GridCode, TrapResult, Pittag, CaptureType, Sex, Weight_uni...
dbl  (6): SamplingYear, NightNumber, TrapNumber, AgeClass, Weight, BodyCondi...
dtm  (1): TrapDate

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Cleaning Data

Renaming Categories

Looking at the [Codebook](#), the variables `Vaccinations`, `ReproductiveStatus` and `BodyCondition` are coded in interesting ways. In `Vaccinations`, missing values (NA) are considered as not vaccinated; therefore, changing the missing to “NV” will ensure that they are correctly classified. In `ReproductiveStatus`, the missing values (NA) indicate unknown reproductive status. In `BodyCondition`, the numerical values 1, 2, 3, 4, and 5, represent a condition. To make the condition easier, we will convert them to the categories. Run the code below to alter them:

```
df <- df |> mutate( # Will change the variables in the df data set
  Vaccinations = case_when(is.na(Vaccinations) ~ "NV", .default = Vaccinations), # Reclassifies Missing
```

```

ReproductiveStatus = case_when(is.na(ReproductiveStatus) ~ "UN", .default = ReproductiveStatus), # Rec
BodyCondition = case_when( # Begins to modify BodyCondition
  as.character(BodyCondition) == "1" ~ "emaciated", # Changes 1 to emaciated
  as.character(BodyCondition) == "2" ~ "thin", # Changes 2 to thin
  as.character(BodyCondition) == "3" ~ "healthy wild", # Changes 3 to healthy wild
  as.character(BodyCondition) == "4" ~ "extra fat reserves", # Changes 4 to Extra Fat Reserve
  as.character(BodyCondition) == "5" ~ "extreme fat reserves" # Changes 5 to Extreme Fat Reserves
)
)

```

Missing Values

Several variables in the data set may be missing. Using the `summary()`, we can determine which variables have missing values by looking at the “NA’s” category.

```
summary(df)
```

Island	SamplingYear	GridCode	
Length:4975	Min. :2014	Length:4975	
Class :character	1st Qu.:2014	Class :character	
Mode :character	Median :2015	Mode :character	
	Mean :2015		
	3rd Qu.:2016		
	Max. :2016		

TrapDate	NightNumber	TrapNumber	
Min. :2014-07-16 00:00:00	Min. :1.000	Min. : 1.00	
1st Qu.:2014-08-24 00:00:00	1st Qu.:2.000	1st Qu.: 4.00	
Median :2015-08-12 00:00:00	Median :3.000	Median : 8.00	
Mean :2015-08-25 01:31:27	Mean :3.392	Mean : 18.87	
3rd Qu.:2016-08-18 00:00:00	3rd Qu.:5.000	3rd Qu.: 11.00	
Max. :2016-11-30 00:00:00	Max. :6.000	Max. :108.00	

TrapResult	Pittag	CaptureType	Sex
Length:4975	Length:4975	Length:4975	Length:4975
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

AgeClass	Weight	Weight_units	BodyCondition
Min. :0.000	Min. : 0.72	Length:4975	Length:4975
1st Qu.:1.000	1st Qu.: 1.73	Class :character	Class :character
Median :1.000	Median : 2.02	Mode :character	Mode :character
Mean :1.557	Mean : 49.49		
3rd Qu.:3.000	3rd Qu.: 2.30		
Max. :4.000	Max. :910.00		
NA's :2840	NA's :2808		
ReproductiveStatus	Vaccinations		
Length:4975	Length:4975		
Class :character	Class :character		

```
Mode :character Mode :character
```

We notice that the variables `AgeClass`, `Weight`, and `BodyCondition` have missing values. **For this analysis, we will remove the missing values (NA).**

```
df <- drop_na(df, AgeClass, Weight, BodyCondition) # Removes the missing values (NA) using the variables A
```

After cleaning the data, this should result in 1,967 recorded foxes!

```
nrow(df)
```

```
[1] 1967
```

Fox Weight Analysis

Imagine you're a field biologist tracking animal health across the remote Santa Rosa and San Miguel Islands. You've noticed that some animals seem heavier in certain years—but is it just your imagination, or is there a real trend?

The `Weight` variable contains information of the Fox's weight in kilograms. Let's start by calculating the mean weight and standard deviation of weight using `mean()` and `sd()` functions:

```
mean_weight<-mean(df$Weight)
```

```
sd_weight<-sd(df$Weight)
```

```
mean_weight
```

```
[1] 1.940275
```

```
sd_weight #printing the values
```

```
[1] 0.4140872
```

We see that mean weight of a fox is about 1.9 kg (which is about 4 pounds) with standard deviation about .41 kg.

We use the `table()` and `prop.table()` functions to get the frequencies and proportions for the `Sex`.

```
count_df <- table(df$Sex) # Using the 'Sex' variable from the `df` data set, we count the frequencies of e
```

```
count_df # Printing out the contents of "rs_df"
```

```
Female  Male
  923    1044
```

Using the `table()` function, we can see that there were 923 female and 1044 male Foxes captured.

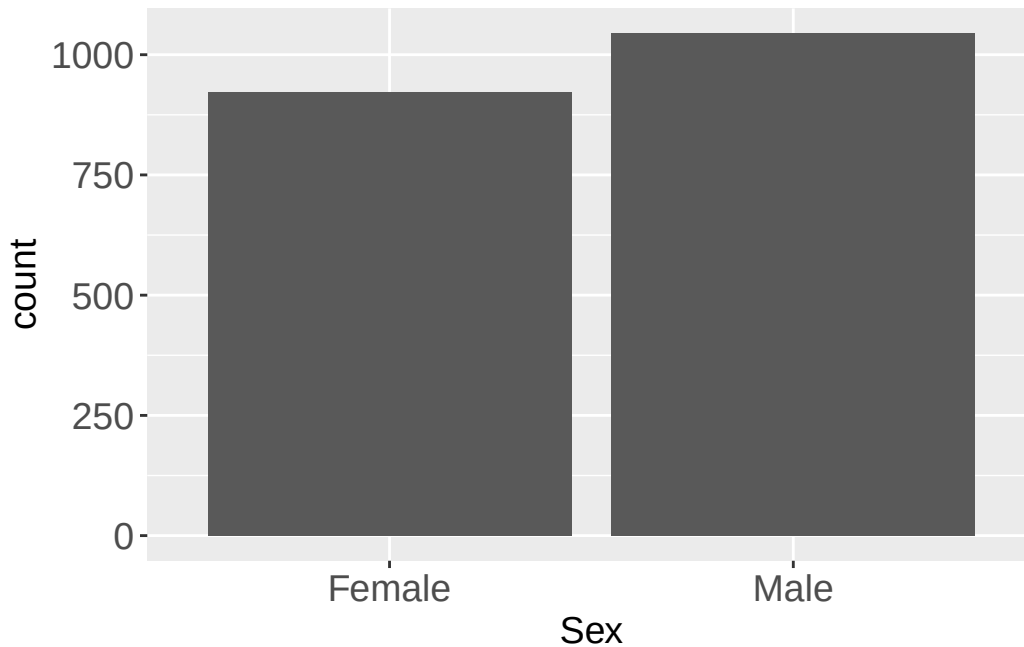
```
prop.table(count_df) # Computing the Proportions of "rs_df"
```

```
Female  Male
0.4692425 0.5307575
```

The `prop.table()` function shows the respective percentages: there are slightly more male Foxes (about 53%) vs. female Foxes (about 47%).

We can visualize the data using the `ggplot` functions. Namely we build a bar plot that shows those above counts visually (with more Male than Female foxes):

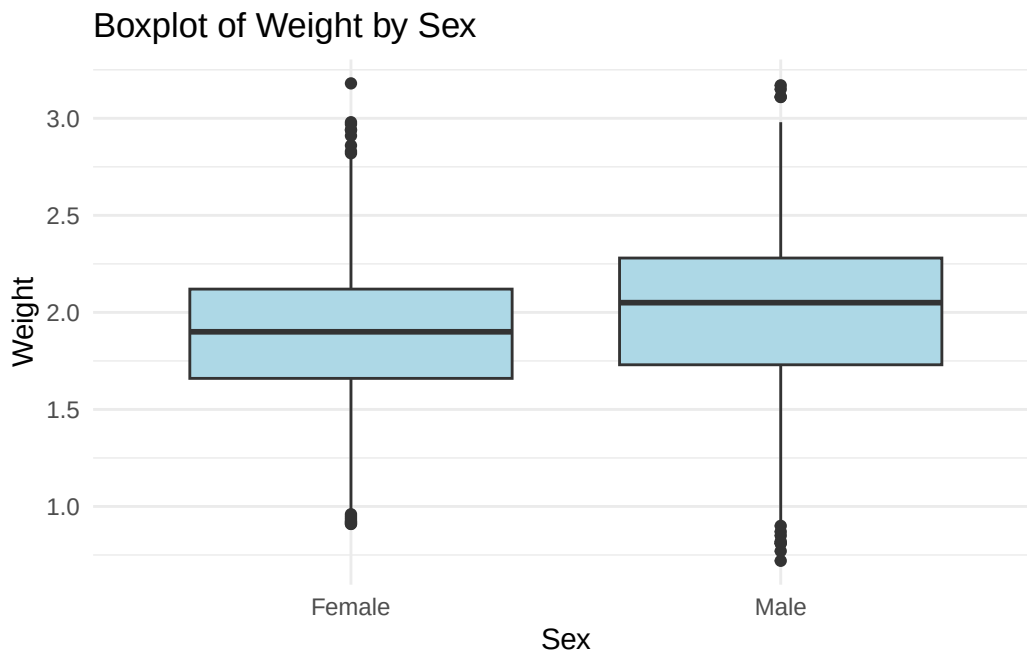
```
ggplot(df) + # Setting up the data to create a plot.  
  geom_bar(aes(Sex)) + # Creating a bar chart based on the variable "ReproductiveStatus"  
  theme(axis.title = element_text(size = 14), axis.text = element_text(size = 14))
```



Weight vs Sex

Let's visualize the variable `Weight` and `Sex` with a boxplot using `ggplot`:

```
ggplot(df, aes(x = Sex, y = Weight)) +  
  geom_boxplot(fill = "lightblue") +  
  labs(  
    title = "Boxplot of Weight by Sex",  
    x = "Sex",  
    y = "Weight"  
  ) + theme_minimal()
```



From the box plot we see that male Foxes tend to be slightly *heavier* than female Foxes.

We can use `group_by()` function to separate male from female population and calculate means and standard deviations separately:

```
weight_summary <- df %>%
  group_by(Sex) %>%
  summarise(
    mean_weight = mean(Weight),
    sd_weight = sd(Weight),
    .groups = "drop"
  )
weight_summary
```

A tibble: 2 x 3

	Sex	mean_weight	sd_weight
	<chr>	<dbl>	<dbl>
1	Female	1.88	0.386
2	Male	1.99	0.431

This table confirms that male Foxes are slightly heavier, with mean weight of about 1.99 kg, than female Foxes, whose mean weight is about 1.88 kg. Standard deviations are also slightly higher for male Foxes.

Weight vs Sex over time

To see weight vs sex trend over time we can use `group_by()` and `ggplot` functions again:

```
weight_summary_yearly <- df %>%
  group_by(Sex, SamplingYear) %>%
  summarise(
    mean_weight = mean(Weight, na.rm = TRUE),
    sd_weight = sd(Weight, na.rm = TRUE),
```

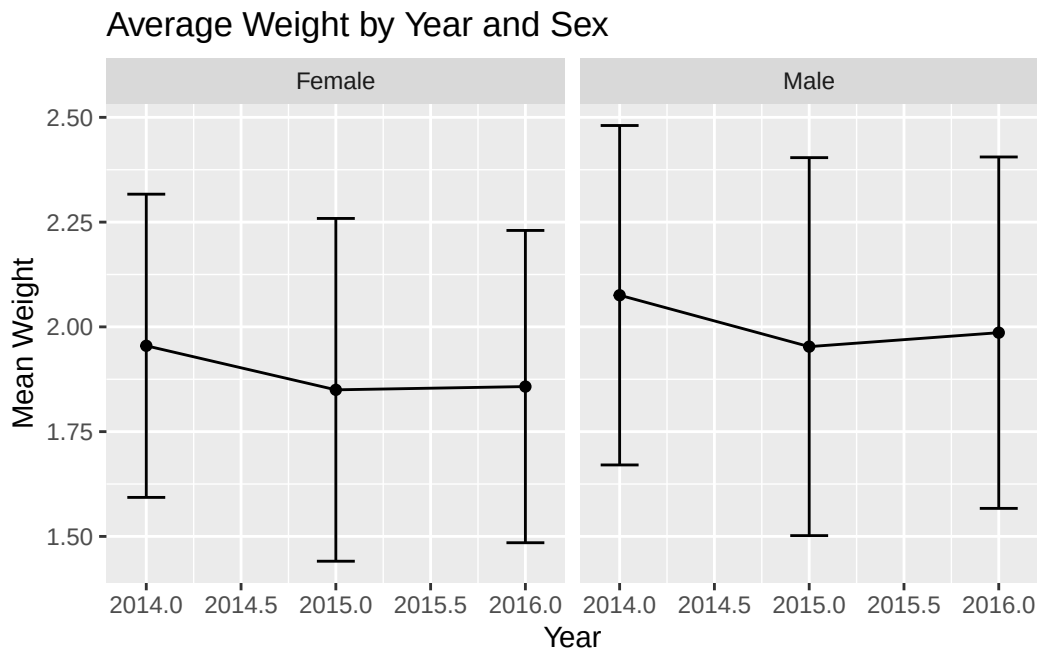
```
.groups = "drop"
) #group the data by Sex and Year and calculate mean and standard deviation of weight
weight_summary_yearly
```

```
# A tibble: 6 x 4
```

	Sex	SamplingYear	mean_weight	sd_weight
	<chr>	<dbl>	<dbl>	<dbl>
1	Female	2014	1.95	0.362
2	Female	2015	1.85	0.409
3	Female	2016	1.86	0.373
4	Male	2014	2.08	0.405
5	Male	2015	1.95	0.451
6	Male	2016	1.99	0.419

From the table we see a downward trend in weight for both females and males across years. Let's plot it on the graph:

```
ggplot(weight_summary_yearly, aes(x = SamplingYear, y = mean_weight)) +
  geom_line() +
  geom_point() +
  geom_errorbar(aes(ymin = mean_weight - sd_weight,
                    ymax = mean_weight + sd_weight), width = 0.2) +
  facet_wrap(~ Sex) +
  labs(
    title = "Average Weight by Year and Sex",
    y = "Mean Weight",
    x = "Year"
  )
)
```



[enhanced jigsaw, title=Weight vs Sex over time by Island, left=2mm, breakable, bottomrule=.15mm, rightrule=.15mm, colframe=quarto-callout-note-color-frame, colback=white, leftrule=.75mm, opacitybackti-


```
tle=0.6, tilerule=0mm, colbacktitle=quarto-callout-note-color!10!white, coltitle=black, bottomtitle=1mm,
opacityback=0, toptitle=1mm, toprule=.15mm, arc=.35mm]
```

Here is a first question for you to answer: do the foxes on different islands have the same weight? Is there an island where the foxes are **significantly** heavier than others? To conduct such analysis, you can filter by different **Islands**:

```
#Santa Rosa Fox population
```

```
SR_df<-filter(df,Island=="SRI")
```

```
#San Miguel Fox population
```

```
SM_df<-filter(df,Island=="SMI")
```

Questions to Think About

```
[enhanced jigsaw, title=Vaccinations, left=2mm, breakable, bottomrule=.15mm, rightrule=.15mm,
colframe=quarto-callout-tip-color-frame, colback=white, leftrule=.75mm, opacitybacktitle=0.6, ti-
tlerule=0mm, colbacktitle=quarto-callout-tip-color!10!white, coltitle=black, bottomtitle=1mm, opaci-
tyback=0, toptitle=1mm, toprule=.15mm, arc=.35mm]
```

Do vaccinated foxes have a different weight compared to unvaccinated ones? To answer this, study the relationship between **Weight** and **Vaccinations**

```
[enhanced jigsaw, title=Body Condition, left=2mm, breakable, bottomrule=.15mm, rightrule=.15mm,
colframe=quarto-callout-warning-color-frame, colback=white, leftrule=.75mm, opacitybacktitle=0.6, ti-
tlerule=0mm, colbacktitle=quarto-callout-warning-color!10!white, coltitle=black, bottomtitle=1mm, opaci-
tyback=0, toptitle=1mm, toprule=.15mm, arc=.35mm]
```

The dataset has a variable called **BodyCondition**. What proportion of foxes are well nourished? Do some islands have a larger proportion of “fat” foxes than others? How does **BodyCondition** connect with **Weight**? For the latter, you could look into scatter plots.

Fox Reproductive Status Analysis

The **ReproductiveStatus** variable contains information of the Fox’s reproductive status when captures. This data is recorded as “N” (Not actively reproductive), “L” (Lactating), “SL” (Signs of Lactating), and “TD” (Testes Distended).

We will use the **table()** and **prop.table()** function to get the frequencies and proportions for the **ReproductiveStatus**.

```
rs_df <- table(df$ReproductiveStatus) # Using the 'ReproductiveStatus' variable from the `df` data set, we
rs_df # Printing out the contents of "rs_df"
```

```

L    SL    TD    UN
18   244  1030  675
```

Using the **table()** function, we can see that there are 3 common reproductive statuses by Fox: “L” at 18, “SL” at 244, “TD” at 1030, and “UN” at 675.

```
prop.table(rs_df) # Computing the Proportions of "rs_df"
```

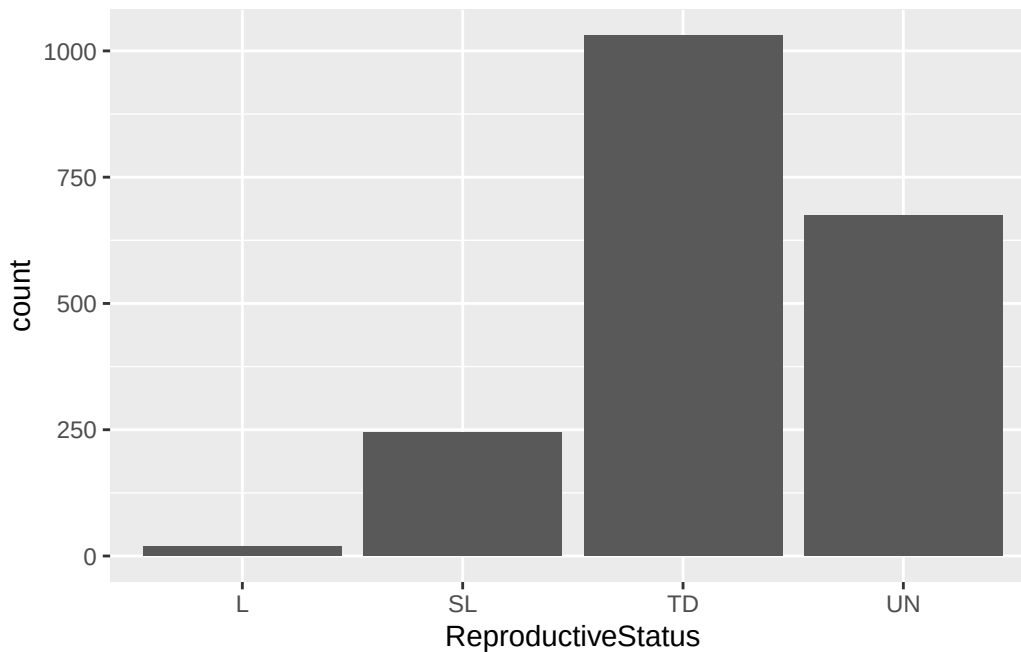
```

L          SL          TD          UN
```

```
0.009150991 0.124046772 0.523640061 0.343162176
```

Using the `prop.table()` function, we can see that the most common status is “TD” at 52.4% and the least common is “L” at the 0.9%. Both “SL” and “UN” represent 12.4% and 34.3% of the data, respectively. In barplot format we have:

```
ggplot(df) + # Setting up the data to create a plot.  
  geom_bar(aes(ReproductiveStatus)) # Creating a bar chart based on the variable "ReproductiveStatus"
```



Sex and Reproductive Status

Let's compare the frequencies between the variables `Sex` and `Reproductive Status`. We will be using the same `table()` function:

```
xy_df <- table(df$ReproductiveStatus, df$Sex)  
# Use the variables "ReproductiveStatus" and "Sex" from the "df" data set  
# Use the table function to compute the crosstabs  
# Store the results in the xy_df object  
  
xy_df # Print results out
```

	Female	Male
L	18	0
SL	244	0
TD	0	1030
UN	661	14

The results show that certain categorical combinations between `ReproductiveStatus` and `Sex` are 0, which is to be expected. The category “UN” is found in both Male and Female foxes. Looking at Female foxes, The most common type is “UN”, followed by “SL”. For Male foxes, the most common type was “TD”.

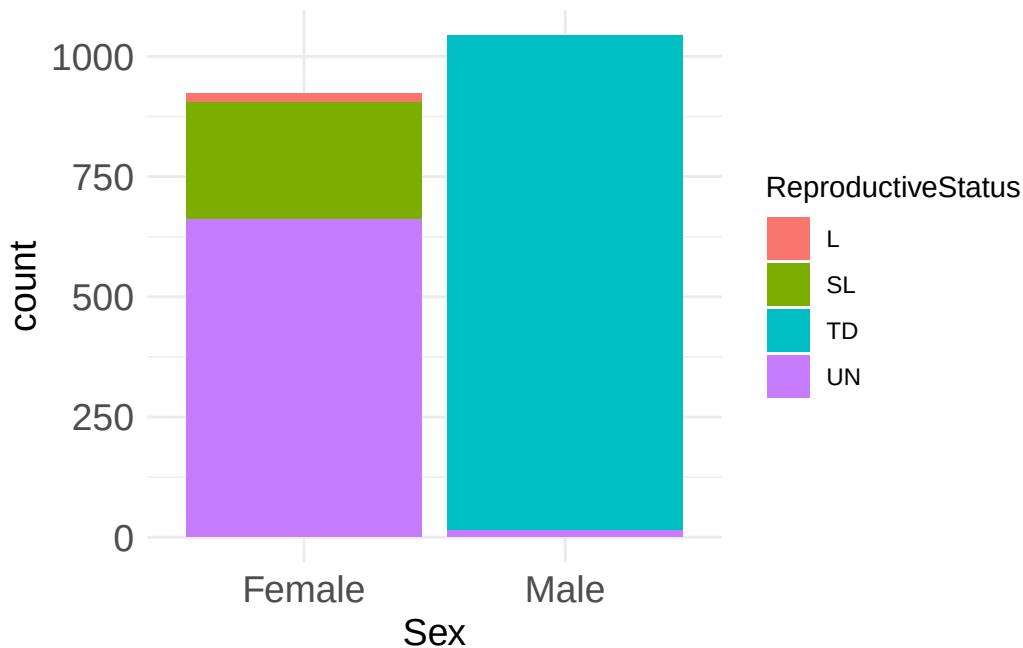
Let's find the proportions for each the combination of the variables using the `prop.table()` function.

```
prop.table(xy_df)
```

	Female	Male
L	0.009150991	0.000000000
SL	0.124046772	0.000000000
TD	0.000000000	0.523640061
UN	0.336044738	0.007117438

We can see that 52.4% of the data are Male and “TD” foxes, 33.6% of the data are Female and “UN” foxes, and 12.4% of the data are Female and “SL” foxes. To visualize how the variable **ReproductiveStatus** and **Sex** related to each other we use a stacked bar plot:

```
ggplot(df) +  
  geom_bar(aes(Sex, fill = ReproductiveStatus)) +  
  theme_minimal() +  
  theme(axis.title = element_text(size = 14), axis.text = element_text(size = 14))
```



The plot indicate which categories are most dominant in each sex. We can see that the most dominant reproductive status in Male foxes is testes distended. For Female foxes, we can see that the most common reproductive status is Unknown.

Challenges: Questions to try out

[enhanced jigsaw, title=Age Class, left=2mm, breakable, bottomrule=.15mm, rightrule=.15mm, colframe=quarto-callout-note-color-frame, colback=white, leftrule=.75mm, opacitybacktitle=0.6, titlerule=0mm, colbacktitle=quarto-callout-note-color!10!white, coltitle=black, bottomtitle=1mm, opacityback=0, toptitle=1mm, toprule=.15mm, arc=.35mm]

How does **ReproductiveStatus** relate to **AgeClass** (how old is the fox), to **Vaccinations**, and to **Capture Type**.

[enhanced jigsaw, title=Trap Results, left=2mm, breakable, bottomrule=.15mm, rightrule=.15mm, colframe=quarto-callout-note-color-frame, colback=white, leftrule=.75mm, opacitybacktitle=0.6, titlerule=0mm, colbacktitle=quarto-callout-note-color!10!white, coltitle=black, bottomtitle=1mm, opacityback=0, toptitle=1mm, toprule=.15mm, arc=.35mm]

How many skunks were caught each year? Are there more skunks on some island than others? (Look at **TrapResult**. Note that some skunks were caught but escaped!)

[enhanced jigsaw, title=Fox Ages, left=2mm, breakable, bottomrule=.15mm, rightrule=.15mm, colframe=quarto-callout-note-color-frame, colback=white, leftrule=.75mm, opacitybacktitle=0.6, titlerule=0mm, colbacktitle=quarto-callout-note-color!10!white, coltitle=black, bottomtitle=1mm, opacityback=0, toptitle=1mm, toprule=.15mm, arc=.35mm]

What is the Age distribution of the foxes? How many foxes are less than 1 year old? Does this vary by year or by island? (Look at **AgeClass**. Notice that it's a numeric variable)

[enhanced jigsaw, title=Tagging, left=2mm, breakable, bottomrule=.15mm, rightrule=.15mm, colframe=quarto-callout-note-color-frame, colback=white, leftrule=.75mm, opacitybacktitle=0.6, titlerule=0mm, colbacktitle=quarto-callout-note-color!10!white, coltitle=black, bottomtitle=1mm, opacityback=0, toptitle=1mm, toprule=.15mm, arc=.35mm]

Each fox gets a unique Pit Tag. How many different foxes have been tagged? If you look at 'CaptureType' you can also analyze how many foxes have been observed multiple times.